EE227 – Digital Logic Design

Lecture 23

Outline

- Analysis with T – Flip Flop
- Mealy and Moore State Machines
- State Reduction
- State Assignment
- Quiz 5

Analysis with T Flip Flop (1/3)

- The analysis with T flip flop is same as the analysis with JK flip flop
- Let us suppose that a circuit with the following T flip flop input equations is given

  $T_A = Bx \quad T_B = x \qquad y = AB$

Here:

- T_A and T_B shows that there are two T flip flops
- A and B are present states of flips flops
- x is input
- y is output

Analysis with T Flip Flop (2/3)

- For JK flip flop the characteristic equations are given as

  $A(t + 1) = T'A + TA' = (T \oplus A)$

  $B(t + 1) = T'B + TB' = (T \oplus B)$

  $T_A = Bx \qquad T_B = x$

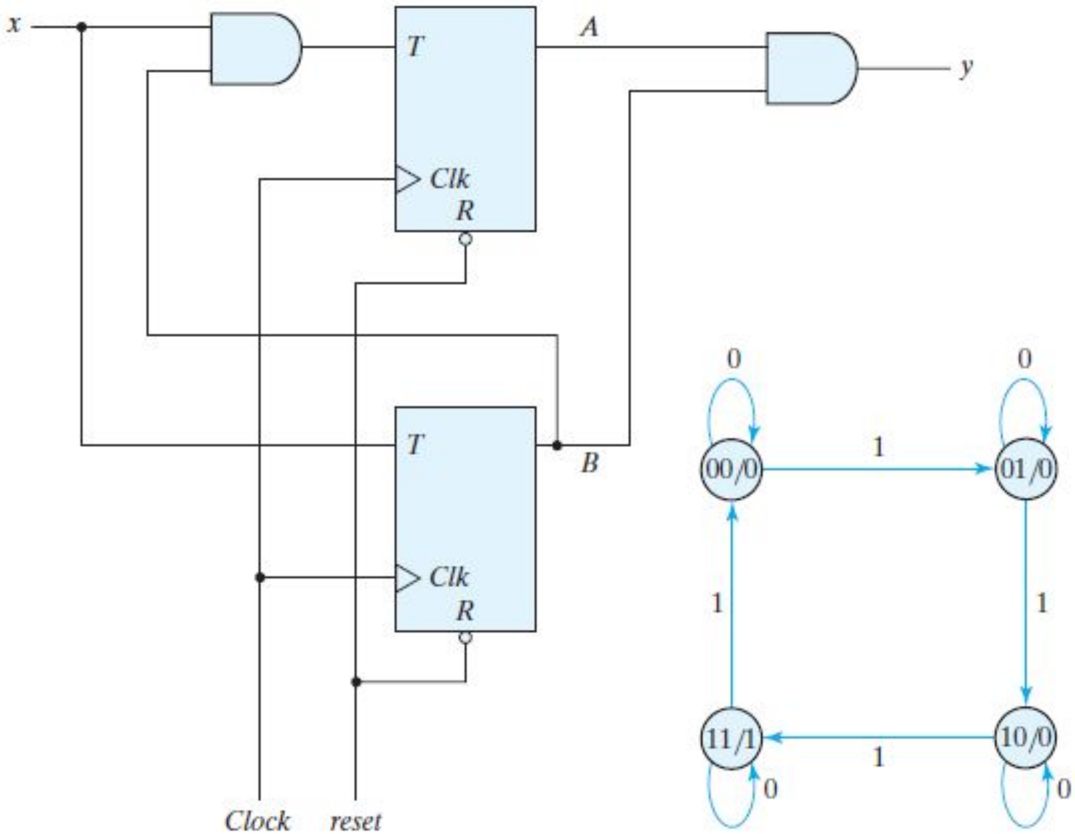
- Substitute the values of T_A and T_B, we get

  $A(t + 1) = (Bx \oplus A)$

  $B(t + 1) = x \oplus B$

- The above two equations are state equations

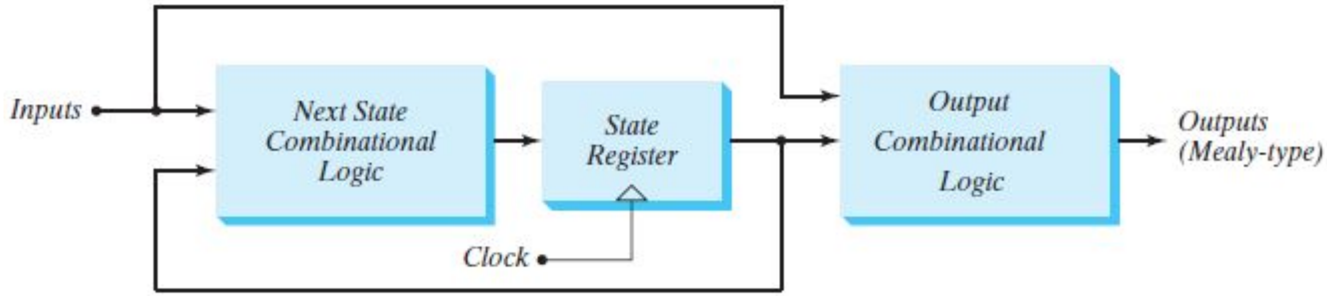
Analysis with T Flip Flop (3/3)



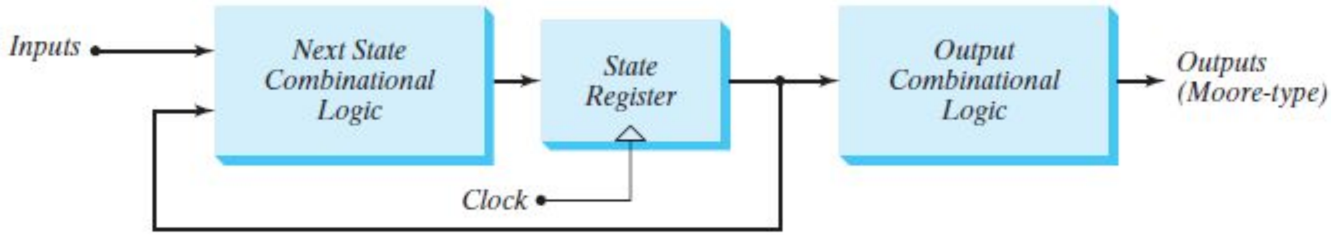
Present State		Input	Next State		Output
A	**B**	**x**	**A**	**B**	**y**
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	0	1	0
0	1	1	1	0	0
1	0	0	1	0	0
1	0	1	1	1	0
1	1	0	1	1	1
1	1	1	0	0	1

Mealy and Moore Machine (1/2)

- Mealy Model
  - The output is a function of both the present state and the input



- Moore Model
  - The output is a function of only the present state



Mealy and Moore Machine (2/2)

- The two models of a sequential circuit are commonly referred to as a finite state machine, abbreviated FSM
- In a Moore model, the outputs of the sequential circuit are synchronized with the clock, because they depend only on flip-flop outputs that are synchronized with the clock
- The output of the Mealy machine is the value that is present immediately before the active edge of the clock

State Reduction

- The reduction in the number of flip-flops in a sequential circuit is referred to as the state-reduction problem
- State-reduction algorithms are concerned with procedures for reducing the number of states in a state table, while keeping the external input–output requirements unchanged
- Since m flip-flops produce 2^m states, a reduction in the number of states may (or may not) result in a reduction in the number of flip-flops
- An unpredictable effect in reducing the number of flip-flops is that sometimes the equivalent circuit (with fewer flip-flops) may require more combinational gates to realize its next state and output logic

Steps to Solve State Reduction Problem

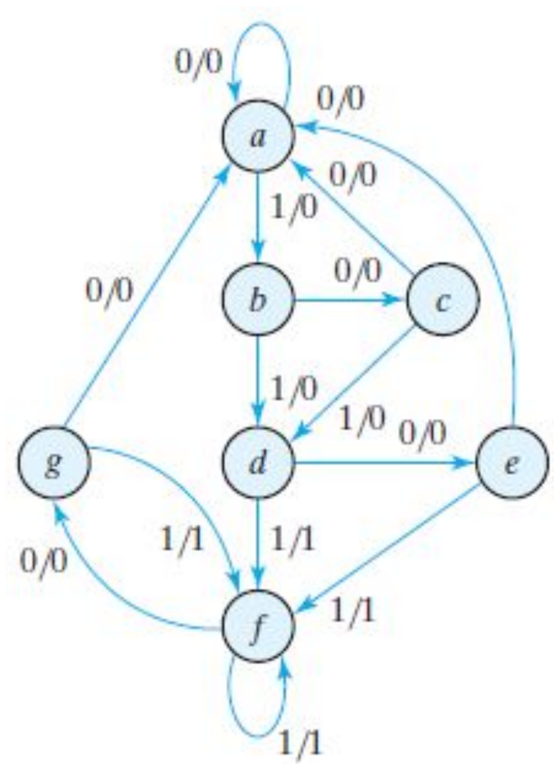
1. Obtain the state table from the given state diagram
2. Find the equivalent states in the state table

   Two states are said to be equivalent if, for each member of the set of inputs, they give exactly the same output and send the circuit either to the same state or to an equivalent state

3. When two states are equivalent, remove one of them and replace the removed state with equivalent state in other columns
4. Continue this step until there are no equivalent states

State Reduction (Example)

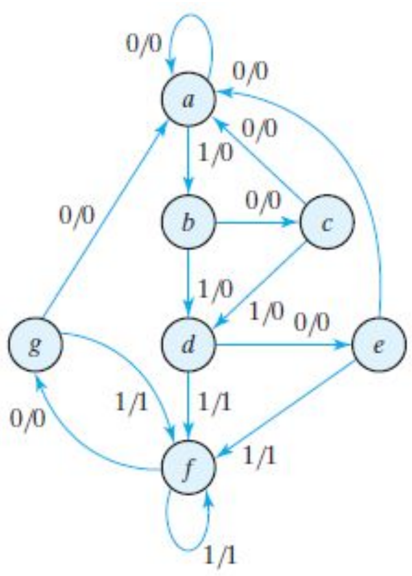
- Reduce the following state diagram to minimum possible states



State Reduction (Example)

- Step 1
  - Obtain the state table

Present State	Next State $x = 0$	Next State $x = 1$	Output $x = 0$	Output $x = 1$
a	a	b	0	0
b	c	d	0	0
c	a	d	0	0
d	e	f	0	1
e	a	f	0	1
f	g	f	0	1
g	a	f	0	1

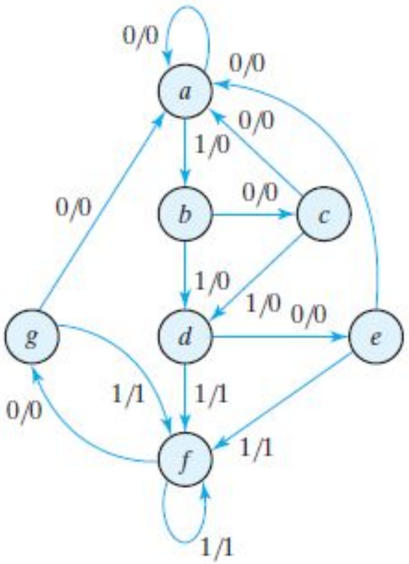


state	a	a	b	c	d	e	f	f	g	f	g	a
input	0	1	0	1	0	1	1	0	1	0	0	
output	0	0	0	0	0	1	1	0	1	0	0	

State Reduction (Example)

- Step 2
  - Find the equivalent states
  - State e and state g are equivalent

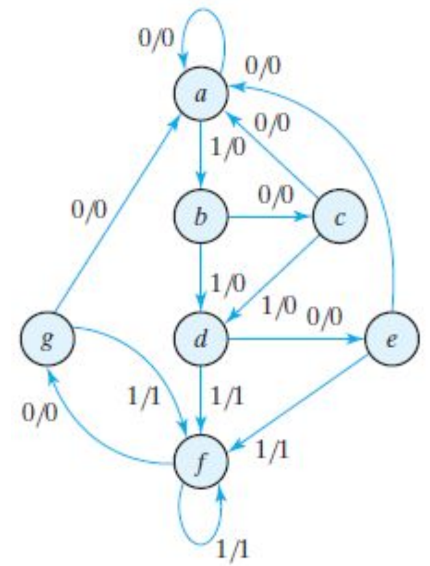
Present State	Next State $x = 0$	Next State $x = 1$	Output $x = 0$	Output $x = 1$
a	a	b	0	0
b	c	d	0	0
c	a	d	0	0
d	e	f	0	1
e	a	f	0	1
f	g	f	0	1
g	a	f	0	1



State Reduction (Example)

- Step 3
  - Remove one of the equivalent states and replace the removed state with equivalent state in other columns
  - Lets remove state “g”, and replace “g” with “e”

	Next State		Output	
Present State	**$x = 0$**	**$x = 1$**	**$x = 0$**	**$x = 1$**
a	a	b	0	0
b	c	d	0	0
c	a	d	0	0
d	e	f	0	1
e	a	f	0	1
f	e	f	0	1



State Reduction (Example)

- Step 4
  - Repeat step 2 and 3 until there are no equivalent states
  - State d and state f are equivalent
  - So remove state f, and replace f with d, everywhere
  - There are no further equivalent states

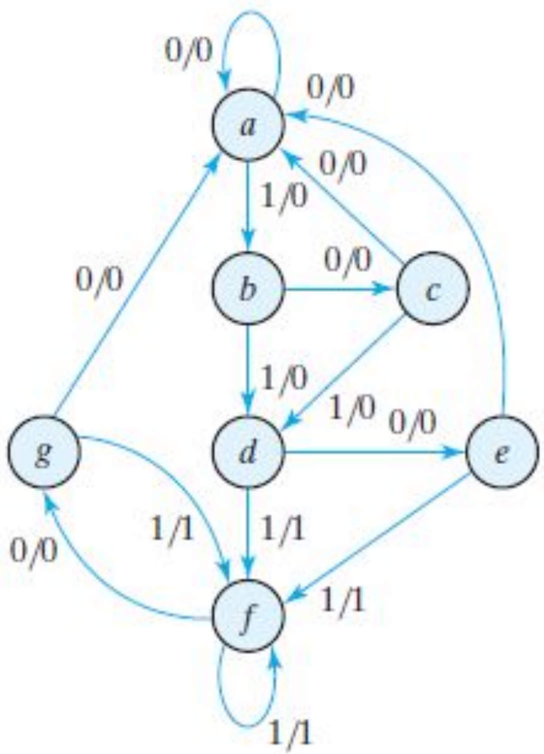
Present State	Next State $x=0$	Next State $x=1$	Output $x=0$	Output $x=1$
a	*a*	*b*	0	0
b	*c*	*d*	0	0
c	*a*	*d*	0	0
d	*e*	*f*	0	1
e	*a*	*f*	0	1
f	*e*	*f*	0	1

Reduced State Table

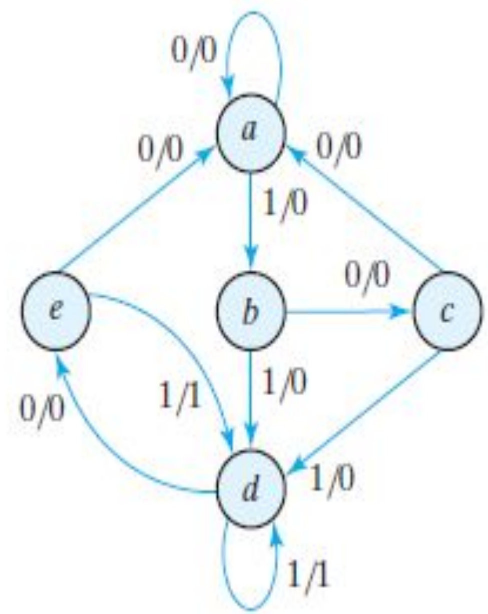
Present State	Next State $x=0$	Next State $x=1$	Output $x=0$	Output $x=1$
a	*a*	*b*	0	0
b	*c*	*d*	0	0
c	*a*	*d*	0	0
d	*e*	*d*	0	1
e	*a*	*d*	0	1

State Reduction (Example)

Given State Diagram



Reduced State Diagram



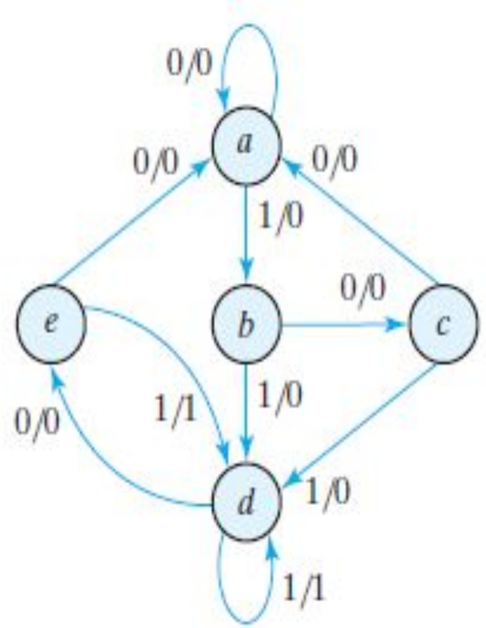
State Assignment

- In order to design a sequential circuit with physical components, it is necessary to assign unique coded binary values to the states
- For a circuit with m states, the codes must contain n bits, where $2^n \geq m$
- For example if there are 5 states in a circuit, then we can assign them binary codes from 000 to 111
- Three of these codes will be unused and will be treated as Don't Care
- We will treat them as Don't Care because the Don't Care conditions often help us in obtaining a simple circuit

Methods of Assignment (1/2)

- Binary
  - The simplest way to code the states is to use the integers in binary counting order
- Gray Code
  - Another assignment may be a gray code assignment where only one bit changes from one state to another
- One – Hot
  - This configuration uses as many bits as there are states in the circuit
  - At any given time, only one bit is equal to 1 while all others are kept at 0

Methods of Assignment (2/2)



State Diagram

State	Assignment 1, Binary	Assignment 2, Gray Code	Assignment 3, One-Hot
a	000	000	00001
b	001	001	00010
c	010	011	00100
d	011	010	01000
e	100	110	10000

Possible Assignment of States